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United States Patent
Kind Code
Date of Patent
Inventor(s)

12405762
B2
September 02, 2025
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System and method for displaying an online electronic document

Abstract

A system for displaying an online electronic document includes a document server, a notification server and a remote conference server. The remote conference server is configured to establish a conference call connection between a first electronic device and a second electronic device. The first electronic device and the second electronic device respectively download from the document server and display an electronic document. After one page of the electronic document displayed on the first electronic device is switched to another page, the first electronic device generates a page turning indication and transmits the page turning indication to the notification server via the document server. The notification server transmits the page turning indication to the second electronic device, and one page of the electronic document displayed on the second electronic device is switched to another page in response to the page turning indication.

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Family ID: 93120992

Appl. No.: 18/641506

Filed: April 22, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240354046 A1	Oct. 24, 2024

Related U.S. Application Data

us-provisional-application US 63460700 20230420

Publication Classification

Int. Cl.: G06F3/14 (20060101); H04L9/40 (20220101); H04L65/1069 (20220101)

U.S. Cl.:

CPC G06F3/1454 (20130101); H04L63/0853 (20130101); H04L65/1069 (20130101);

Field of Classification Search

CPC: G06F (21/32); G06F (21/645); G06F (3/0304); G06F (3/03545); G06F (3/04883); G06F (40/171); G06F (40/174); G06F (2211/004); G06F (3/1454); G06Q (10/10); G06Q (40/03); G06Q (50/18); H04L (12/1831); H04L (9/3247); H04L (63/0853); H04L (63/10); H04L (63/126); H04L (65/1069); G09G (2358/00)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present disclosure claims the benefit of and priority to U.S. provisional Patent Application Ser. No. 63/460,700 filed on Apr. 20, 2023, entitled “Online Signature System, Method and Electronic Apparatus,” (hereinafter referred to as “the '700 provisional”). The disclosure of the '700 provisional is hereby incorporated fully by reference into the present disclosure.

FIELD

(1) The present disclosure generally relates to a system and method for displaying an electronic document, and more particularly, to a system and method for displaying an online electronic document.

BACKGROUND

(2) With the development of global e-commerce, many document or contract reviews and signature execution are no longer limited to paper documents, and gradually the paper documents have been replaced by electronic documents. However, when an electronic document needs to be reviewed and/or signed by multiple parties online, there is room for improvement in enabling the electronic document to be synchronously displayed and operated (for example, synchronously turning pages) on the display devices of multiple parties in an online conference call.

SUMMARY

(3) The present disclosure provides a system for displaying an online electronic document

including a remote conference server, a document server and a notification server. The remote conference server is configured to establish a conference call connection between a first electronic device and a second electronic device. The document server is configured to provide an electronic document for the first electronic device and the second electronic device to download. The electronic document includes at least a first page and a second page. After the electronic document is downloaded by the first electronic device and the second electronic device, respectively, a first page of the electronic document is synchronously displayed on a first display of the first electronic device and a second display of the second electronic device. The first page displayed on the first display is switched to the second page through a page turning operation to cause the first electronic device to generate a page turning indication and transmit the page turning indication to the document server. The document server is configured to receive the page turning indication from the first electronic device and transmit the page turning indication to the notification server. The notification server is configured to receive the page turning indication from the document server and transmit the page turning indication to the second electronic device. After the page turning indication is received by the second electronic device, the first page displayed on the second display is switched to the second page in response to the page turning indication.

(4) In the system according to an embodiment of the present disclosure, the second page has a signature area. After the first page displayed on the second display is switched to the second page in response to the page turning indication, the signature area is displayed on the second display for receiving a signature object. The signature object is generated based on an input received by the second electronic device and is presented on the signature area displayed on the second display. The document server is configured to receive the signature object from the second electronic device and transmit the signature object to the notification server. After the notification server receives the signature object from the document server, the notification server transmits the signature object to the first electronic device. After the signature object is received by the first electronic device, the signature object is presented on the signature area of the second page displayed on the first display, such that the signature object is synchronously presented on the signature area displayed on the first display and on the signature area displayed on the second display.

(5) In the system according to an embodiment of the present disclosure, the signature object is a static signature image, and the static signature image is encoded into an image format string by the second electronic device. The signature object received and transmitted by the document server to the notification server is the image format string. The signature object received and transmitted by the notification server to the first electronic device is the image format string. After the image format string is received by the first electronic device, the image format string is decoded into the static signature image and presented on the signature area of the second page displayed on the first display.

(6) In the system according to another embodiment of the present disclosure, the signature object is a dynamic signature image, and the dynamic signature image is encoded into an image format string array by the second electronic device. The signature object received by the document server from the second electronic device is the image format string array, the document server is further configured to convert the image format string array into a video format string array, and the signature object transmitted by the document server to the notification server is the video format string array. The signature object received and transmitted by the notification server to the first electronic device is the video format string array. After the video format string array is received by the first electronic device, the video format string array is decoded into a dynamic signature video and presented on the signature area of the second page displayed on the first display.

(7) In the system according to another embodiment of the present disclosure, the signature object includes a static signature image and a dynamic signature image, the static signature image is encoded into an image format string by the second electronic device, and the dynamic signature image is encoded into an image format string array by the second electronic device. The signature

object received by the document server from the second electronic device includes the image format string and the image format string array, the document server is further configured to convert the image format string array into a video format string array, and the signature object transmitted by the document server to the notification server includes the image format string and the video format string array. The signature object received and transmitted by the notification server to the first electronic device includes the image format string and the video format string array. After the image format string and the video format string array are received by the first electronic device, the image format string is decoded into the static signature image and the video format string array is decoded into a dynamic signature video. The decoded static signature image and the dynamic signature video are sequentially presented on the signature area of the second page displayed on the first display.

(8) The present disclosure provides a method for displaying an online electronic document, and the method includes: establishing, by a remote conference server, a conference call connection between a first electronic device and a second electronic device; providing, by a document server, an electronic document including at least a first page and a second page to be downloaded by the first electronic device and the second electronic device; displaying the first page of the electronic document on a first display of the first electronic device and a second display of the second electronic device synchronously after the electronic document is downloaded by the first electronic device and the second electronic device, respectively; switching the first page displayed on the first display to the second page through a page turning operation to generate a page turning indication; receiving, by the document server, the page turning indication and transmitting the page turning indication to a notification server; receiving, by the notification server, the page turning indication from the document server and transmitting the page turning indication to the second electronic device; and switching the first page displayed on the second display to the second page in response to the page turning indication received from the notification server.

(9) The method according to the present embodiment of the present disclosure further includes: displaying a signature area of the second page on the second display after the first page displayed on the second display is switched to the second page; receiving, by the second electronic device, an input to generate a signature object and presenting the signature object on the signature area displayed on the second display; receiving, by the document server, the signature object from the second electronic device and transmitting the signature object to the notification server; receiving, by the notification server, the signature object from the document server and transmitting the signature object to the first electronic device; and presenting the signature object on the signature area of the second page displayed on the first display after the signature object is received by the first electronic device.

(10) In the method according to an embodiment of the present disclosure, the signature object is a static signature image, and the static signature image is encoded into an image format string. The signature object received and transmitted by the document server to the notification server is the image format string. The signature object received and transmitted by the notification server to the first electronic device is the image format string. Presenting the signature object on the signature area of the second page displayed on the first display further includes decoding the image format string into the static signature image and presenting the static signature image on the signature area of the second page displayed on the first display.

(11) In the method according to another embodiment of the present disclosure, the signature object is a dynamic signature image, and the dynamic signature image is encoded into an image format string array. The signature object received by the document server from the second electronic device is the image format string array, the image format string array is converted by the document server into a video format string array, and the signature object transmitted by the document server to the notification server is the video format string array. The signature object received and transmitted by the notification server to the first electronic device is the video format string array.

Presenting the signature object on the signature area of the second page displayed on the first display further includes decoding the video format string array into a dynamic signature video and presenting the dynamic signature video on the signature area of the second page displayed on the first display.

(12) In the method according to another embodiment of the present disclosure, the signature object includes a static signature image and a dynamic signature image, the static signature image is encoded into an image format string by the second electronic device, and the dynamic signature image is encoded into an image format string array by the second electronic device. The signature object received by the document server from the second electronic device includes the image format string and the image format string array, the image format string array is converted into a video format string array by the document server, and the signature object transmitted by the document server to the notification server includes the image format string and the video format string array. The signature object received and transmitted by the notification server to the first electronic device includes the image format string and the video format string array. Presenting the signature object on the signature area of the second page displayed on the first display further includes: decoding the image format string into the static signature image and decoding the video format string array into a dynamic signature video; and presenting the decoded static signature image and the dynamic signature video sequentially on the signature area of the second page displayed on the first display.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to make the above and other objects, features, advantages and embodiments of the present disclosure more readily understood, the accompanying drawings are described as follows:
- (2) FIG. 1 is a schematic diagram illustrating an architecture of a system for displaying an online electronic document according to an embodiment of the present disclosure.
- (3) FIG. 2 is a schematic diagram of signaling and channels among devices of the system for displaying an online electronic document according to an embodiment of the present disclosure.
- (4) FIG. 3 is a flowchart of a method for displaying an online electronic document according to a first embodiment of the present disclosure.
- (5) FIG. 4A is a schematic diagram showing a page of an electronic document presented synchronously on a user interface of a host APP and a user interface of a client APP according to an embodiment of the present disclosure.
- (6) FIG. 4B is a schematic diagram showing another page (without including the signature object) of an electronic document presented on a user interface of a host APP and a user interface of a client APP synchronously according to an embodiment of the present disclosure.
- (7) FIG. 4C is a schematic diagram showing another page (including the signature object) of an electronic document presented on a user interface of a host APP and a user interface of a client APP synchronously according to an embodiment of the present disclosure.
- (8) FIG. 5 is a flowchart of a method for displaying an online electronic document according to a second embodiment of the present disclosure.
- (9) FIG. 6 is a flowchart of a method for displaying an online electronic document according to a third embodiment of the present disclosure.
- (10) FIG. 7 is a flowchart of a method for displaying an online electronic document according to a fourth embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(11) In the present disclosure, “a,” “an,” and “the” may refer to a singular form or a plural form, unless an article is specifically restricted to be a singular form in the context.

(12) In addition, as used herein, the terms “comprise/comprising,” “include/including,” “have/having,” and the like are open-ended terms that imply the inclusion of the disclosed features, elements and/or components, but do not preclude the presence or addition of one or more other features, elements, components, and/or groups thereof.

(13) The term “coupled” used in the present disclosure may indicate that two or more elements or devices are in direct physical contact with each other or in indirect physical contact with each other, and may also indicate that two or more elements or devices cooperate or interact with each other.

(14) Furthermore, the ordinal terms (such as “first,” “second,” “third,” and the like) used in the present disclosure and claims are used to modify an element itself and do not imply any priority or any order of one element over another element, or do not imply a chronological order of steps of a method performed, but are used only as symbols to distinguish a claimed element having a particular name from another element having the same name.

(15) The spirit of the present disclosure will be clearly illustrated with drawings and detailed descriptions below. After understanding the embodiments of the present disclosure, those skilled in the art with ordinary knowledge can make modifications and variations based on the technologies taught in the present disclosure without departing from the spirit and scope of the present disclosure.

(16) FIG. 1 is a schematic diagram illustrating an architecture of a system **100** for displaying an online electronic document according to an embodiment of the present disclosure. Referring to FIG. 1, the system **100** for displaying an online electronic document includes a remote conference server **110**, a document server **120** and a notification server **130**. The remote conference server **110**, the document server **120**, and the notification server **130** are electronic devices having computing and networking functions and provide services through a network **140**. The remote conference server **110** is configured to provide a conference call connection service. The document server **120** is configured to provide an electronic document service. The notification server **130** is configured to provide an operation record and notification service for electronic documents.

(17) The remote conference server **110** establishes a conference call connection between a first electronic device **150** and a second electronic device **160** through the network **140**, such that the first electronic device **150** and the second electronic device **160** may have a video conversation or a voice conversation. In addition, in other embodiments, the remote conference server **110** can also establish a conference call connection between the first electronic device **150** and a plurality of second electronic devices **160** at the same time.

(18) In the present embodiment, the system **100** for displaying an online electronic document provides two types of applications (APPs), i.e., a host APP **151** and a client APP **161**, for users with different access permissions to use services provided by the system **100**. In the present embodiment, the services provided by the system **100** includes synchronously turning a page of an online electronic document and/or executing a signature on the online electronic document. The first electronic device **150** is a device used by a host (e.g., a salesperson providing an electronic document) and has a host APP **151** installed thereon. The second electronic device **160** is a device used by a user (e.g., a client reading and/or executing a signature on an electronic document) and has a client APP **161** installed thereon.

(19) The first electronic device **150** includes a processor **152**, a storage **153**, a communication component **154**, and a display **155**. The processor **152** is electrically coupled to the storage **153**, the communication component **154**, and the display **155**. The storage **153** may store the host APP **151**.

(20) The second electronic device **160** includes a processor **162**, a storage **163**, a communication component **164**, and a display **165**. The processor **162** is electrically coupled to the memory **163**, the communication component **164**, and the display **165**. The storage **163** may store the client app **161**.

(21) The processors **152**, **162** may be, for example, a central processing unit (CPU), a physics processing unit (PPU), a programmable microprocessor, an embedded controlling chip, a digital

signal processor (DSP), an application specific integrated circuit (ASIC), or that like.

(22) The storages **153**, **163** may be any type of fixed or removable random access memory (RAM), read-only memory (ROM), flash memory, hard disk, or other similar device, or a combination of these devices.

(23) The communication components **154**, **164** may be chips or circuits that employ the local area network (LAN) technology, the wireless LAN (WLAN) technology, or the mobile communication technology. The LAN technology, for example, is Ethernet. The WLAN technology, for example, is Wi-Fi. The mobile communication technology, for example, is the global system for mobile communications (GSM) technology, the third-generation (3G) technology, the fourth-generation (4G) technology, the fifth-generation (5G) technology, or the like.

(24) The displays **155** and **165** are, for example, a liquid crystal display (LCD), a plasma display (PDP), an organic Light-Emitting diode (OLED) display, or the like.

(25) The host APP **151** is installed and stored in the storage **153** of the first electronic device **150** and may be executed by the processor **152**. When the host APP **151** is executed by the processor **152**, an operation interface of the host APP **151** is displayed on the display **155**. Similarly, the client APP **161** is installed and stored in the storage **163** of the second electronic device **160** and may be executed by the processor **162**. When the client APP **161** is executed by the processor **162**, an operation interface of the client APP **161** is displayed on the display **165**.

(26) In addition, the remote conference server **110**, the document server **120**, and the notification server **130** each have a processor, a storage, and a communication component (not shown but similar to the above-mentioned processors **152**, **162**, storages **153**, **163**, and communication components **154**, **164**).

(27) FIG. 2 is a schematic diagram of signaling and channels among devices of the system for displaying an online electronic document according to an embodiment of the present disclosure. Referring to FIG. 2, the first electronic device **150** schedules a conference call with the remote conference server **110** through the host APP **151**. For example, the first electronic device **150** transmits a conference request S1 to the remote conference server **110** through the host APP **151**. Then, the remote conference server **110** transmits a reservation notification S2 to the client APP **161** of the second electronic device **160** according to the conference request S1 transmitted by the first electronic device **150**. Thereafter, the client APP **161** of the second electronic device **160** transmits a confirmation notification S3 to the remote conference server **110**. Then, the remote conference server **110** transmits a connection information S4 and connection information S5 to the second electronic device **160** and the first electronic device **150**, respectively. Finally, the first electronic device **150** establishes a conference call connection S6 with the client APP **161** of the second electronic device **160** through the host APP **151**. During the conference call connection S6, the document server **120** may provide a communication channel S7 to transmit data or control signals to the second electronic device **160** or receive data or control signals from the second electronic device **160** through the communication channel S7. In an embodiment of the present disclosure, the document server **120** may establish the communication channel S7 according to at least one task request signal received from the second electronic device **160** and establish a communication channel S8 according to at least one task request signal received from the first electronic device **150**. During the conference call connection S6, the document server **120** may provide another communication channel S8 to transmit data or control signals to the first electronic device **150** or receive data or control signals from the first electronic device **150** through the communication channel S8. In addition, during the conference call connection S6, a communication channel S9 is established between the document server **120** and the notification server **130** for transmitting data or control signals between the document server **120** and the notification server **130**. Furthermore, during the conference call connection S6, the notification server **130** may provide a communication channel S10 for transmitting data or control signals to the first electronic device **150** or receiving data or control signals from the first electronic device **150**.

through the communication channel S10. During the conference call connection S6, the notification server **130** may further provide a communication channel S11 for transmitting data or control signals to the second electronic device **160** or receiving data or control signals from the second electronic device **160** through the communication channel S11. In an embodiment of the present disclosure, the notification server **130** may establish the communication channel S10 according to at least one channel request signal received from the first electronic device **150** and establish the communication channel S11 according to at least one channel request signal received from the second electronic device **160**.

(28) FIG. **3** is a flowchart of a method for displaying an online electronic document according to a first embodiment of the present disclosure. Referring to FIG. **1** to FIG. **3**, in step S202, a conference call connection S6 between the first electronic device **150** and the second electronic device **160** is established by the remote conference server **110**. Specifically, the first electronic device **150** and the second electronic device **160** are connected to the network **140** via the communication component **154** and the communication component **164**, respectively, to establish by the remote conference server **110** the conference call connection S6 between the first electronic device **150** and the second electronic device **160**. In the present embodiment, the conference call connection S6 may be a video telephone connection or a voice telephone connection, such that the host using the first electronic device **150** (e.g., a salesperson providing an electronic document) and the user using the second electronic device **160** (e.g., a client reading and/or executing a signature on the electronic document) can communicate and have a conversation through the conference call connection S6.

(29) In step S204, the document server **120** provides an electronic document for the first electronic device **150** and the second electronic device **160** to download. For example, after the conference call connection S6 is established, the first electronic device **150** may download the electronic document from the document server **120** through the communication channel S8, and the second electronic device **160** may download the electronic document from the document server **120** through the communication channel S7.

(30) In step S206, after the first electronic device **150** and the second electronic device **160** each download the electronic document from the document server **120**, the electronic document is displayed synchronously on the display **155** of the first electronic device **150** and the display **165** of the second electronic device **160**. For example, after the first electronic device **150** and the second electronic device **160** respectively download an electronic document **170** (as shown in FIG. **4A**) from the document server **120**, the electronic document **170** is presented on a user interface **156** of the host APP **151** displayed on the display **155** of the first electronic device **150** and is also presented on a user interface **166** of the client APP **161** displayed on the display **165** of the second electronic device **160**.

(31) FIG. **4A** is a schematic diagram showing the downloaded electronic document **170** presented synchronously on the user interface **156** of the host APP **151** and the user interface **166** of the client APP **161** according to an embodiment of the present disclosure. In the present embodiment, the electronic document **170** may be, for example, an insurance contract and include two pages. After the first electronic device **150** and the second electronic device **160** respectively download the electronic document **170**, at least one page (e.g., the first page **170a**) of the electronic document **170** is respectively presented on the user interface **156** of the host APP **151** displayed on the display **155** and on the user interface **166** of the client APP **161** displayed on the display **165**. In the present embodiment, the user interface **156** of the host APP **151** includes a document display area **156a**, a video window **156b**, a previous page button **156c**, a next page button **156d**, and a plurality of function buttons A1, B1, C1, and D1. The document display area **156a** is used to present at least one page (e.g., the first page **170a**) of the electronic document **170**, the video window **156b** is used to display a video image of a user (e.g., a client reading and/or executing signature on the electronic document **170**) of the second electronic device **160** connected to the first electronic device **150** for

a video call, the previous page button **156c** is used to turn a current page of the electronic document **170** presented on the user interface **156** to a previous page, and the next page button **156d** is used to turn the current page of the electronic document **170** presented on the user interface **156** to a next page. The plurality of function buttons A1, B1, C1, and D1 may be software buttons related to other functions, such as, but not limited to, a document open button, a document save button, an end call button, and a start call button. In addition, the user interface **166** of the client APP **161** includes a document display area **166a**, a video window **166b**, a previous page button **166c**, a next page button **166d**, and a plurality of function buttons A2, B2, C2, and D2. The document display area **166a** is used to present at least one page (e.g., the first page **170a**) of the electronic document **170**, the video window **166b** is used to display a video image of a host (e.g., a salesperson providing the electronic document **170**) of the first electronic device **150** connected to the second electronic device **160** for the video call, the previous page button **166c** is used to turn a current page of the electronic document **170** presented on the user interface **166** to a previous page, and the next page button **166d** is used to turn the current page of the electronic document **170** presented on the user interface **166** to a next page. The plurality of function buttons A2, B2, C2, and D2 may be software buttons related to other functions, such as, but not limited to, a document open button, a document save button, an end call button, a start call button, and a signature completion button.

(32) In step **S208**, a current page of the electronic document displayed on the display **155** is switched to another page through a page turning operation to cause the processor **152** of the first electronic device **150** to generate a page turning indication and transmit, via the communication component **154**, the page turning indication to the document server **120** through the communication channel **S8**. For example, after the host of the first electronic device **150** explains contents listed on the first page **170a** of the electronic document **170** to the user of the second electronic device **160** through the conference call connection established by the remote conference server **110**, the host may turn the first page **170a** (as shown in FIG. **4A**) of the electronic document **170** to the second page **170b** (as shown in FIG. **4B**) through a page turning operation. In the present embodiment, the page turning operation may be completed by selecting the next page button **156d** on the user interface **156** via an input device (e.g., a mouse, a keyboard, or a touch panel), but is not limited thereto. After the next page button **156d** is selected, the first page **170a** of the electronic document **170** presented on the user interface **156** is turned and switched to the second page **170b** (as shown in FIG. **4B**), and the processor **152** of the first electronic device **150** may generate a page turning indication and transmit, via the communication component **154**, the page turning indication to the document server **120** through the communication channel **S8**. In an embodiment of the present disclosure, the page turning indication may include page turning information, such as the page number “2” of the second page **170b** of the electronic document **170**, to indicate that the electronic document **170** on the user interface **156** is turned and switched to the second page **170b**.

(33) In step **S210**, the document server **120** receives the page turning indication from the first electronic device **150** and transmits the page turning indication to the notification server **130** through the communication channel **S9**.

(34) In step **S212**, the notification server **130** receives the page turning indication from the document server **120** and transmits the page turning indication to the second electronic device **160** through the communication channel **S11**. In another embodiment of the present disclosure, when the remote conference server **110** simultaneously establishes a conference call connection between the first electronic device **150** and the plurality of second electronic devices **160**, the notification server **130** transmits the page turning indication to the plurality of second electronic devices **160** simultaneously. Each of the plurality of second electronic devices **160** performs the same action after receiving the page turning indication. Therefore, the second electronic device **160** mentioned in the following step **S214** may refer to a single second electronic device **160** or each of the plurality of second electronic devices **160**.

(35) In step **S214**, after the page turning indication is received by the second electronic device **160**,

a current page displayed on the display **165** is switched to another page in response to the page turning indication. For example, after the second electronic device **160** receives, via the communication component **164**, the page turning indication from the notification server **130**, the first page **170a** of the electronic document **170** presented on the user interface **166** is turned and switched to the second page **170b** (as shown in FIG. 4B) in response to the page turning indication, such that the first page **170a** of the electronic document **170** presented on the user interface **156** and the first page **170a** of the electronic document **170** presented on the user interface **166** (as shown in FIG. 4A) may be turned and switched to the second page **170b** (as shown in FIG. 4B) through the same page turning operation (e.g., a selection of the next page button **156d**).

(36) In the present embodiment, after the first page **170a** of the electronic document **170** presented on the user interface **156** and the first page **170a** of the electronic document **170** presented on the user interface **166** are both switched to the second page **170b** (as shown in FIG. 4B), the host of the first electronic device **150** may turn the second page **170b** of the electronic document **170** back to the first page **170a** through another page turning operation (e.g., a selection of the previous page button **156c**). In addition, the user of the second electronic device **160** may turn the second page **170b** of the electronic document **170** back to the first page **170a** through another page turning operation (e.g., a selection of the previous page button **166c**).

(37) In a second embodiment of the present disclosure, the method illustrated in FIG. 3 may further include steps **S216**, **S218**, **S220**, and **S222** (as shown in FIG. 5). The host of the first electronic device **150** may turn the second page **170b** of the electronic document **170** back to the first page **170a** through steps **S216**, **S218**, **S220**, and **S222**.

(38) In step **S216**, a current page of the electronic document displayed on the display **155** is switched to a previous page through a second page turning operation to cause the processor **152** of the first electronic device **150** to generate a second page turning indication and transmit, via the communication component **154**, the second page turning indication to the document server **120** through the communication channel **S8**. For example, the host may turn the second page **170b** (as shown in FIG. 4B) of the electronic document **170** back to the first page **170a** (as shown in FIG. 4A) through the second page turning operation. In the present embodiment, the second page turning operation may be completed by selecting the previous page button **156c** on the user interface **156** via the above-mentioned input device (e.g., a mouse, a keyboard, or a touch panel), but is not limited thereto. After the previous page button **156c** is selected, the second page **170b** of the electronic document **170** presented on the user interface **156** is turned and switched back to the first page **170a** (as shown in FIG. 4A), and the processor **152** of the first electronic device **150** may generate a second page turning indication and transmit, via the communication component **154**, the second page turning indication to the document server **120** through the communication channel **S8**. In an embodiment of the present disclosure, the second page turning indication may include page turning information, such as the page number “1” of the first page **170a** of the electronic document **170**, to indicate that the electronic document **170** on the user interface **156** is turned and switched back to the first page **170a**.

(39) In step **S218**, the document server **120** receives the second page turning indication from the first electronic device **150** and transmits the second page turning indication to the notification server **130** through the communication channel **S9**.

(40) In step **S220**, the notification server **130** receives the second page turning indication from the document server **120** and transmits the second page turning indication to the second electronic device **160** through the communication channel **S11**. In another embodiment of the present disclosure, when the remote conference server **110** simultaneously establishes a conference call connection between the first electronic device **150** and the plurality of second electronic devices **160**, the notification server **130** transmits the second page turning indication to the plurality of second electronic devices **160** simultaneously. Each of the plurality of second electronic devices **160** may perform the same action after receiving the second page turning indication. Therefore, the

second electronic device **160** mentioned in the following step **S222** may refer to a single second electronic device **160** or each of the plurality of second electronic devices **160**.

(41) In step **S222**, after the second page turning indication is received by the second electronic device **160**, a current page displayed on the display **165** is switched back to a previous page in response to the second page turning indication. For example, after the second electronic device **160** receives, via the communication component **164**, the second page turning indication from the notification server **130**, the second page **170b** of the electronic document **170** presented on the user interface **166** is turned and switched back to the first page **170a** (as shown in FIG. 4A) in response to the second page turning indication, such that the second page **170b** of the electronic document **170** presented on the user interface **156** and the second page **170b** of the electronic document **170** presented on the user interface **166** (as shown in FIG. 4B) may be turned and switched back to the first page **170a** (as shown in FIG. 4A) through the same page turning operation (e.g., a selection of the previous page button **156c**).

(42) In a third embodiment of the present disclosure, the method illustrated in FIG. 3 may further include steps **S224**, **S226**, **S228**, and **S230** (as shown in FIG. 6). The user of the second electronic device **160** may turn the second page **170b** of the electronic document **170** back to the first page **170a** through steps **S224**, **S226**, **S228**, and **S230**.

(43) In step **S224**, a current page of the electronic document displayed on the display **165** is switched to a previous page through a third page turning operation to cause the processor **162** of the second electronic device **160** to generate a third page turning indication and transmit, via the communication component **164**, the third page turning indication to the document server **120** through the communication channel **S7**. For example, the user may turn the second page **170b** (as shown in FIG. 4B) of the electronic document **170** back to the first page **170a** (as shown in FIG. 4A) through the third page turning operation. In the present embodiment, the third page turning operation may be completed by selecting the previous page button **166c** on the user interface **166** via an input device (e.g., a mouse, a keyboard, or a touch panel), but is not limited thereto. After the previous page button **166c** is selected, the second page **170b** of the electronic document **170** presented on the user interface **166** is turned and switched back to the first page **170a** (as shown in FIG. 4A), and the processor **162** of the second electronic device **160** may generate a third page turning indication and transmit, via the communication component **164**, the third page turning indication to the document server **120** through the communication channel **S7**. In an embodiment of the present disclosure, the third page turning indication may include page turning information, such as the page number “1” of the first page **170a** of the electronic document **170**, to indicate that the electronic document **170** on the user interface **166** is turned and switched back to the first page **170a**.

(44) In step **S226**, the document server **120** receives the third page turning indication from the second electronic device **160** and transmits the third page turning indication to the notification server **130**.

(45) In step **S228**, the notification server **130** receives the third page turning indication from the document server **120** and transmits the third page turning indication to the first electronic device **150** through the communication channel **S10**. In another embodiment of the present disclosure, when the remote conference server **110** simultaneously establishes a conference call connection between the first electronic device **150** and the plurality of second electronic devices **160**, the notification server **130** transmits the third page turning indication to the plurality of second electronic devices **160** simultaneously. The action performed by each of the plurality of second electronic devices **160** after receiving the third page turning indication may be the same with the action performed by the first electronic device **150** after receiving the third page turning indication. Therefore, in the following step **S230**, the first electronic device **150**, the communication component **154**, the display **155**, and the user interface **156** can also respectively represent the second electronic devices **160** that receives the third page turning indication, the communication

component **164**, the display **165**, and the user interface **166**.

(46) In step **S230**, after the third page turning indication is received by the first electronic device **150**, a current page displayed on the display **155** is switched back to a previous page in response to the third page turning indication. For example, after the first electronic device **150** receives, via the communication component **154**, the third page turning indication from the notification server **130**, the second page **170b** of the electronic document **170** presented on the user interface **156** is turned and switched back to the first page **170a** (as shown in FIG. **4A**) in response to the third page turning indication, such that the second page **170b** of the electronic document **170** presented on the user interface **166** and the second page **170b** of the electronic document **170** presented on the user interface **156** (as shown in FIG. **4B**) may be turned and switched back to the first page **170a** (as shown in FIG. **4A**) through the same page turning operation (e.g., a selection of the previous page button **166c**).

(47) Referring to FIG. **4B**, the second page **170b** of the electronic document **170** has a signature area **172**. The user of the second electronic device **160** may execute a signature on the signature area **172** after reviewing the contents of the electronic document **170**.

(48) In a fourth embodiment of the present disclosure, the method illustrated in FIG. **3** may further include steps **S232**, **S234**, **S236**, **S238**, and **S240** (as shown in FIG. **7**). Referring to FIG. **1**, FIG. **2**, FIGS. **4A-4C** and FIG. **7**, after the first page **170a** of the electronic document **170** presented on the user interface **156** and the first page **170a** of the electronic document **170** presented on the user interface **166** are both switched to the second page **170b** (as shown in FIG. **4B**), the user of the second electronic device **160** may perform steps **S232**, **S234**, **S236**, **S238**, and **S240** to execute a signature on the signature area **172** of the second page **170b** of the electronic document **170** to generate a signature object **174**, such that the signature object **174** may be presented synchronously on the electronic document **170** that is presented on the user interface **166** displayed on the display **165** and on the electronic document **170** that is presented on the user interface **156** displayed on the display **155** (as shown in FIG. **4C**).

(49) In step **S232**, after the first page **170a** of the electronic document **170** presented by the user interface **166** displayed on the display **165** is switched to the second page **170b**, the signature area **172** on the second page **170b** is displayed on the display **165** (as shown in FIG. **4B**).

(50) In step **S234**, the second electronic device **160** receives an input to generate the signature object **174** and presents the signature object **174** on the signature area **172** displayed on the display **165** (as shown in FIG. **4C**), and then transmits the signature object **174** to the document server **120** via the communication component **164**. For example, that second electronic device **160** may receive an input via an input device (e.g., a handwriting tablet, a touch panel, or an image capture device) to generate the signature object **174**. In a case that an input is received from a handwriting tablet or a touch panel, the signature object **174** generated is a handwritten signature track (e.g., Joe). In a case that an input is received from an image capture device, the signature object **174** generated is a stamp image or a handwritten signature image. In addition, the second electronic device **160** may transmit, via the communication component **164**, the signature object **174** to the document server **120** through the communication channel **S7**.

(51) In step **S236**, the document server **120** receives the signature object **174** from the second electronic device **160** and transmits the signature object **174** to the notification server **130** through the communication channel **S9**.

(52) In step **S238**, the notification server **130** receives the signature object **174** from the document server **120** and transmits the signature object **174** to the first electronic device **150** through the communication channel **S10**. In another embodiment of the present disclosure, when the remote conference server **110** simultaneously establishes a conference call connection between the first electronic device **150** and the plurality of second electronic devices **160**, the notification server **130** transmits the signature object **174** to the plurality of second electronic devices **160** simultaneously. The action performed by each of the plurality of second electronic devices **160** after receiving the

signature object **174** may be the same with the action performed by the first electronic device **150** after receiving the signature object **174**. Therefore, in the following step **S240**, the first electronic device **150**, the communication component **154**, the display **155**, and the user interface **156** can also respectively represent the second electronic devices **160** that receives the signature object **174**, the communication component **164**, the display **165**, and the user interface **166**.

(53) In step **S240**, after the signature object **174** is received by the first electronic device **150**, the signature object **174** is presented on the signature area **172** of the second page **170b** displayed on the display **155** (as shown in FIG. **4C**). For example, after that first electronic device **150** receives the signature object **174** from the notification server **130** via the communication component **154**, the processor **152** may present the signature object **174** on the signature area **172** of the second page **170b** displayed by the display **155**, such that the signature object **174** may be presented synchronously on the second page **170b** of the electronic document **170** presented on the user interface **166** and on the second page **170b** of the electronic document **170** presented on the user interface **156** through the same signature operation performed by the user (as shown in FIG. **4C**).

(54) In another embodiment of that present disclosure, after the signature object **174** is presented synchronously on the electronic document **170** that is presented on the user interface **166** displayed on the display **165** and on the electronic document **170** that is presented on the user interface **156** displayed on the display **155** (as shown in FIG. **4C**), the user of the second electronic device **160** may select (for example, by using a mouse, touch panel or handwriting tablet) a signature completion button provided on the client APP **161** to transmit to the first electronic device **150** a notification of completing the signature task. For example, when the signature completion button provided on the client APP **161** is selected, the processor **162** may generate a signature completion signal and transmit the signature completion signal to the document server **120** through the communication channel **S7**. The document server **120** may receive the signature completion signal and transmit the signature completion signal to the notification server **130** through the communication channel **S9**. In addition, the document server **120** may also transmit the signature completion signal to the remote conference server **110** to notify the remote conference server **110** that the electronic document **170** has been signed. Then, the notification server **130** receives the signature completion signal from the document server **120** and transmits the signature completion signal to the first electronic device **150** through the communication channel **S10**. Finally, the first electronic device **150** receives the signature completion signal, and the processor **152** generates a notification message to be displayed on the display **155** according to the signature completion signal. In another embodiment of the present disclosure, after the document server **120** receives the signature completion signal, the document server **120** may combine the signature object **174** received in step **S236** with the electronic document **170** to obtain a signed document and store the signed document in a database.

(55) In the present embodiment of the present disclosure, when a user of the second electronic device **160** executes a signature of his or her name (e.g., Joe) on the signature area **172** of the second page **170b** of the electronic document **170** via an input device (e.g., a handwriting tablet or a touch panel), the input device may generate a plurality of signals according to a signature trace of the user on the input device, and the processor **162** may receive the plurality of signals and generate the signature object **174** based on the plurality of signals.

(56) In an embodiment of the present disclosure, the signature object **174** may be a static signature image (e.g., a static image showing a handwritten “Joe”), and the static signature image may be encoded into an image format string (e.g., a Base64 image format string) by the processor **162** of the second electronic device **160**. In step **234**, the second electronic device **160** may transmit, via the communication device **164**, the image format string representing the signature object **174** to the document server **120** through the communication channel **S7**. In step **S236**, the document server **120** receives the image format string and transmits the image format string to the notification server **130** through the communication channel **S9**. In step **S238**, the notification server **130** receives the

image format string from the document server **120** and transmits the image format string to the first electronic device **150** through the communication channel S10. In step S240, after the image format string is received by the first electronic device **150**, the processor **152** may decode the image format string back into the static signature image (e.g., the static image showing the handwritten “Joe”) and present the static signature image on the signature area **172** of the second page **170b** displayed on the display **155** (as shown in FIG. 4C).

(57) In another embodiment of the present disclosure, the signature object **174** may be a dynamic signature image (e.g., a dynamic image showing a handwritten “Joe”), and the dynamic signature image may be encoded into an image format string array (e.g., a Base64 image format string array) by the processor **162** of the second electronic device **160**. The image format string array may consist of a plurality of string (e.g., a plurality of Base64 image format strings), and the plurality of image format strings may respectively represent a plurality of static images generated while the handwritten “Joe” was being signed. For example, the plurality of static images may include a first static image representing “J,” a second static image representing “Jo,” and a third static image representing “Joe” for recording the signature track of Joe's handwritten signature. In step **234**, the second electronic device **160** may transmit, via the communication device **164**, the image format string array representing the signature object **174** to the document server **120** through the communication channel S7. In step S236, the document server **120** receives the image format string array, converts the image format string array into a video format string array (e.g., Base64 video format string array), and transmits the video format string array to the notification server **130** through the communication channel S9. In step S238, the notification server **130** receives the video format string array and transmits the video format string array to the first electronic device **150** through the communication channel S10. In step S240, after the video format string array is received by the first electronic device **150**, the processor **152** may decode the video format string array into a dynamic signature video (e.g., a video showing the handwritten “Joe”) and present the dynamic signature video on the signature area **172** of the second page **170b** displayed on the display **155** (as shown in FIG. 4C). For example, when the dynamic signature video is a video showing the handwritten “Joe” and presented on the signature area **172** of the second page **170b** displayed on the display **155**, the signature track of the handwritten “Joe” is dynamically played. In the embodiment of the present disclosure, the dynamic signature video is a dynamic signature video in MP4 format but is not limited thereto.

(58) In other embodiments of the present disclosure, the signature object **174** may include a static signature image (e.g., a static image showing the handwritten “Joe”) and a dynamic signature image (e.g., a dynamic image showing the handwritten “Joe”). The static signature image may be encoded into an image format string (e.g., a Base64 string) by the processor **162** of the second electronic device **160**, and the dynamic signature image may be encoded into an image format string array (e.g., a Base64 image format string array) by the processor **162** of the second electronic device **160**. In step **234**, the second electronic device **160** may transmit, via the communication device **164**, the image format string and the image format string array representing the signature object **174** to the document server **120** through the communication channel S7. In step S236, the document server **120** receives the image format string and the image format string array from the second electronic device **160** and converts the image format string array into a video format string array (e.g., Base64 video format string array). Then, the document server **120** may sequentially or simultaneously transmit the image format string and the video format string array to the notification server **130** through the communication channel S9. In step S238, the notification server **130** may receive the image format string and the video format string array sequentially or simultaneously and transmit the image format string and the video format string array sequentially or simultaneously to the first electronic device **150** through the communication channel S10. In step S240, after the image format string and the video format string array are sequentially or simultaneously received by the first electronic device **150**, the processor **152** may decode the image

format string back into the static signature image (e.g., the static image showing the handwritten “Joe”) and decode the video format string array into a dynamic signature video (e.g., a video showing the handwritten “Joe”). Afterwards, the decoded static signature image and the dynamic signature video are sequentially presented on the signature area **172** of the second page **170b** displayed on the display **155** (as shown in FIG. **4C**). In an embodiment of the present disclosure, the processor **152** first presents the decoded static signature image on the signature area **172** of the second page **170b** displayed on the display **155** (as shown in FIG. **4C**), and then presents, according to a selection of a function button provided on the host APP **151** and corresponding to a function of displaying the dynamic signature video, the dynamic signature video on the signature area **172** of the second page **170b** displayed on the display **155**, thus replacing the decoded static signature image. In the embodiment of the present disclosure, the dynamic signature video is a dynamic signature video in MP4 format but is not limited thereto.

(59) In summary, with the system and method provided by the present disclosure for displaying an online electronic document, a multiple participants can review and/or execute a signature on the online electronic document together. In addition, when a participant A performs an operation (such as turning a page or executing a signature) on an electronic document, all other participants B can simultaneously see the operation performed by the participant A on the electronic document from the electronic devices they use, thus improving the efficiency of multiple participants reviewing the same electronic document. In addition, in the system and method provided by the present disclosure for displaying an online electronic document, when no operation (such as turning a page or executing a signature) is performed on the electronic document, no image data related to the electronic document is transmitted between each device and each server. Therefore, compared with other technologies that use network streaming to review electronic documents, the system and the method provided by the present disclosure for displaying an online electronic document can effectively save the network bandwidth and effectively reduce the labor and time cost for reviewing the electronic documents.

(60) Although the present disclosure has been disclosed by way of above embodiments, the embodiments are not intended to limit the present disclosure, and those skilled in the art will appreciate that changes and modifications may be made therein as long as those changes and modifications do not deviate from the spirit and the scope of the present disclosure. Therefore, the scope of the present disclosure should be construed according to the definitions in the appended claims.

Claims

1. A system for displaying an online electronic document, comprising: a remote conference server configured to establish a conference call connection between a first electronic device and a second electronic device; a document server; and a notification server, wherein: the document server is configured to provide an electronic document for the first electronic device and the second electronic device to download, and the electronic document comprises at least a first page and a second page, wherein the second page has a signature area; after the electronic document is downloaded by the first electronic device and the second electronic device from the document server, respectively, the first page of the electronic document is synchronously displayed on a first display of the first electronic device and a second display of the second electronic device; the first page displayed on the first display is switched to the second page through a first page turning operation to cause the first electronic device to generate a first page turning indication and transmit the first page turning indication to the document server; the document server is configured to receive the first page turning indication from the first electronic device and transmit the first page turning indication to the notification server; the notification server is configured to receive the first page turning indication from the document server and transmit the first page turning indication to

the second electronic device; after the first page turning indication is received by the second electronic device, the first page displayed on the second display is switched to the second page in response to the first page turning indication; after the first page displayed on the second display is switched to the second page in response to the first page turning indication, the second page displayed on the second display is switched to the first page through a second page turning operation to cause the second electronic device to generate a second page turning indication and transmit the second page turning indication to the document server; the document server is further configured to receive the second page turning indication from the second electronic device and transmit the second page turning indication to the notification server; the notification server is further configured to receive the second page turning indication from the document server and transmit the second page turning indication to the first electronic device; and after the second page turning indication is received by the first electronic device, the second page displayed on the first display is switched to the first page in response to the second page turning indication.

2. The system of claim 1, wherein: after the first page displayed on the second display is switched to the second page in response to the first page turning indication, the signature area is displayed on the second display for receiving a signature object, wherein the signature object is generated based on an input received by the second electronic device; after the signature object is generated, the signature object is presented on the signature area displayed on the second display; the document server is configured to receive the signature object from the second electronic device and transmit the signature object to the notification server; after the notification server receives the signature object from the document server, the notification server transmits the signature object to the first electronic device; and after the signature object is received by the first electronic device, the signature object is presented on the signature area of the second page displayed on the first display, such that the signature object is synchronously presented on the signature area displayed on the first display and on the signature area displayed on the second display.

3. The system of claim 2, wherein: the signature object is a static signature image, and the static signature image is encoded into an image format string by the second electronic device; the signature object received and transmitted by the document server to the notification server is the image format string; the signature object received and transmitted by the notification server to the first electronic device is the image format string; and after the image format string is received by the first electronic device, the image format string is decoded into the static signature image and presented on the signature area of the second page displayed on the first display.

4. The system of claim 2, wherein: the signature object is a dynamic signature image, and the dynamic signature image is encoded into an image format string array by the second electronic device; the signature object received by the document server from the second electronic device is the image format string array, the document server is further configured to convert the image format string array into a video format string array, and the signature object transmitted by the document server to the notification server is the video format string array; the signature object received and transmitted by the notification server to the first electronic device is the video format string array; and after the video format string array is received by the first electronic device, the video format string array is decoded into a dynamic signature video and presented on the signature area of the second page displayed on the first display.

5. The system of claim 2, wherein: the signature object includes a static signature image and a dynamic signature image, the static signature image is encoded into an image format string by the second electronic device, and the dynamic signature image is encoded into an image format string array by the second electronic device; the signature object received by the document server from the second electronic device includes the image format string and the image format string array, the document server is further configured to convert the image format string array into a video format string array, and the signature object transmitted by the document server to the notification server includes the image format string and the video format string array; the signature object received

and transmitted by the notification server to the first electronic device includes the image format string and the video format string array; after the image format string and the video format string array are received by the first electronic device, the image format string is decoded into the static signature image and the video format string array is decoded into a dynamic signature video; and the decoded static signature image and the dynamic signature video are sequentially presented on the signature area of the second page displayed on the first display.

6. A method for displaying an online electronic document, comprising: establishing, by a remote conference server, a conference call connection between a first electronic device and a second electronic device; providing, by a document server, an electronic document including at least a first page and a second page to be downloaded by the first electronic device and the second electronic device, wherein the second page has a signature area; displaying the first page of the electronic document on a first display of the first electronic device and a second display of the second electronic device synchronously after the electronic document is downloaded by the first electronic device and the second electronic device from the document server, respectively; switching the first page displayed on the first display to the second page through a first page turning operation to generate a first page turning indication; receiving, by the document server, the first page turning indication and transmitting the first page turning indication to a notification server; receiving, by the notification server, the first page turning indication from the document server and transmitting the first page turning indication to the second electronic device; switching the first page displayed on the second display to the second page in response to the first page turning indication received from the notification server; after the first page displayed on the second display is switched to the second page in response to the first page turning indication, switching the second page displayed on the second display to the first page through a second page turning operation to generate a second page turning indication; receiving, by the document server, the second page turning indication and transmitting the second page turning indication to the notification server; receiving, by the notification server, the second page turning indication from the document server and transmitting the second page turning indication to the first electronic device; and switching the second page displayed on the first display to the first page in response to the second page turning indication received from the notification server.

7. The method of claim 6, further comprising: displaying the signature area of the second page on the second display after the first page displayed on the second display is switched to the second page; receiving, by the second electronic device, an input to generate a signature object and presenting the signature object on the signature area displayed on the second display; receiving, by the document server, the signature object from the second electronic device and transmitting the signature object to the notification server; receiving, by the notification server, the signature object from the document server and transmitting the signature object to the first electronic device; and presenting the signature object on the signature area of the second page displayed on the first display after the signature object is received by the first electronic device.

8. The method of claim 7, wherein: the signature object is a static signature image, and the static signature image is encoded into an image format string; the signature object received and transmitted by the document server to the notification server is the image format string; the signature object received and transmitted by the notification server to the first electronic device is the image format string; and presenting the signature object on the signature area of the second page displayed on the first display further comprises: decoding the image format string into the static signature image and presenting the static signature image on the signature area of the second page displayed on the first display.

9. The method of claim 7, wherein: the signature object is a dynamic signature image, and the dynamic signature image is encoded into an image format string array; the signature object received by the document server from the second electronic device is the image format string array, the image format string array is converted by the document server into a video format string array,

and the signature object transmitted by the document server to the notification server is the video format string array; the signature object received and transmitted by the notification server to the first electronic device is the video format string array; and presenting the signature object on the signature area of the second page displayed on the first display further comprises: decoding the video format string array into a dynamic signature video and presenting the dynamic signature video on the signature area of the second page displayed on the first display.

10. The method of claim 7, wherein: the signature object includes a static signature image and a dynamic signature image, the static signature image is encoded into an image format string by the second electronic device, and the dynamic signature image is encoded into an image format string array by the second electronic device; the signature object received by the document server from the second electronic device includes the image format string and the image format string array, the image format string array is converted into a video format string array by the document server, and the signature object transmitted by the document server to the notification server includes the image format string and the video format string array; the signature object received and transmitted by the notification server to the first electronic device includes the image format string and the video format string array; and presenting the signature object on the signature area of the second page displayed on the first display further comprises: decoding the image format string into the static signature image and decoding the video format string array into a dynamic signature video; and presenting the decoded static signature image and the dynamic signature video sequentially on the signature area of the second page displayed on the first display.
